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Tutorial Sheet

Stacks and Queues

1. The postfix expression for the infix expression A+B*(C+D)/F+D*E ?

2. The result evaluating the postfix expression 10 5 + 60 6 / * 8 – is

3. Consider a sequence a of elements a0 = 1, a1 = 5, a2 = 7, a3 = 8, a4 = 9, and a5 = 2. The following operations are performed on a stack S and a queue Q, both of which are initially empty.

   I: push the elements of a from a_0 to a_5 in that order into S.

   II: enqueue the elements of a_0 to a_5 to a in that order into Q.

   III: pop an element from S.

   IV: dequeue an element from Q.

   V: pop an element from S.

   VI: dequeue an element from Q.

   VII: dequeue an element from Q and push the same element into S.

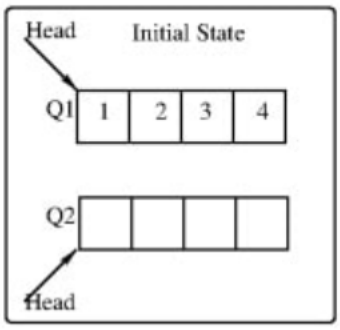
   VIII: Repeat operation VII three times.

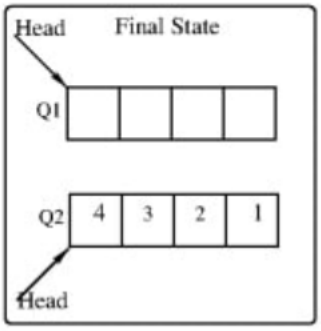
   IX: pop an element from S.

   X: pop an element from S.

   The top element of S after executing the above operations is _____.m

4. Consider the queues Q1 containing four elements and Q2 containing none (shown as the Initial State in the figure). The only operations allowed on these two queues are Enqueue(Q, element) and Dequeue(Q). The minimum number of Enqueue operations on Q1 required to place the elements of Q1 in Q2 in reverse order (shown as the Final State in the figure) without using any additional storage is_____.





5. Suppose a circular queue of capacity (n-1) elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as

array index variables, respectively. Initially, REAR = FRONT = 0. The conditions to detect queue full and queue empty is ?

6. Let Q denote a queue containing sixteen numbers and S be an empty stack. Head(Q) returns the element at the head of the queue Q without removing it from Q. Similarly Top(S) returns the element at the top of S without removing it from S. Consider the algorithm given below.

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while Q is not Empty do
 if S is Empty OR Top(S) ≤ Head (Q) then
   x:= Dequeue (Q);
   Push (S, x);
 else
   x:= Pop(S);
   Enqueue (Q, x);
 end
end
```

The maximum possible number of iterations of the while loop in the algorithm is ?

7. The five items: A, B, C, D, and E are pushed in a stack, one after other starting from A. The stack is popped four items and each element is inserted in a queue. The two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is ?

8. Let S be a stack of size n>=1. Starting with the empty stack, suppose we push the first 5 natural numbers in sequence, and then perform 5 pop operations. Assume that Push and Pop operations take 3 seconds each, and 1 seconds elapse between the end of one such stack operation and the start of the next operation. The average stack-life of an element of this stack is ___.

9. Convert the following infix expression to its equivalent prefix: K+L-M*N+(O^P)*W/U/V*T+Q

10. Which of the following permutations can be obtained in the output (in the same order) using a stack assuming that the input is the sequence 1, 2, 3, 4, 5 in that order?
    a. 3,4,5,1,2
    b. 3,4,5,2,1
    c. 1,5,2,3,4
    d. 5,4,3,1,2